COMPUTER SCIENCE AND SOCIETY

Who I am

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AGENDA

UNIT 1

COMPUTER SCIENCE, SOCIETY, AND THE INFORMATION SOCIETY

On completion of this unit, you will be able to...

- explain the **importance** of computer science to society.
- outline the **development** of computer science and the **Internet**.
- name relevant **organizations** that help shape the field of computer science and society.

1. Explain what is meant by the “**Information Society**”!

2. Briefly outline the **history of the Internet** until today!

3. Name one **female computing pioneer!**

- US: “Computer Science” since the 1940s
- In Germany: introduced as a course of study in the late 1960s
- Information and communication technology (ICT) created by computer scientists shapes all aspects of society and the economy!

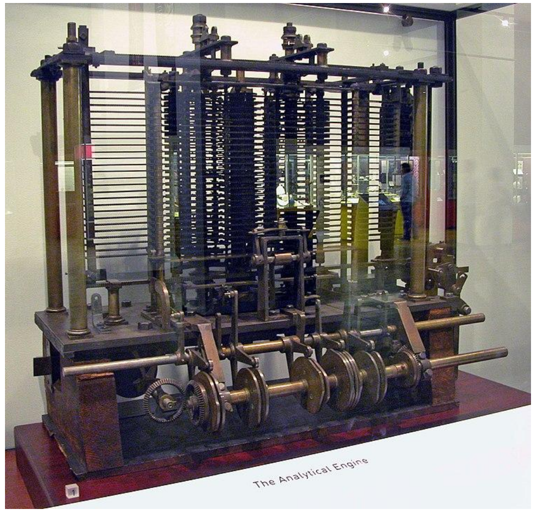
- Technical layer: algorithms, code, data structures, networks, hardware, databases.
- Human layer: users, workers, students, citizens, communities.
- Institutional layer: companies, universities, governments, schools, regulators.
- Social layer: norms, culture, power relations, inclusion, trust and risk.

WHY SOCIETY MATTERS FOR COMPUTER SCIENCE

- Scale: one application can reach billions of users.
- Opacity: algorithms can influence decisions without being visible.
- Dependence: work, education, and public services increasingly rely on digital systems.
- Power: platforms can mediate markets, communication, and knowledge access.
- Responsibility: technical choices can create unequal outcomes.

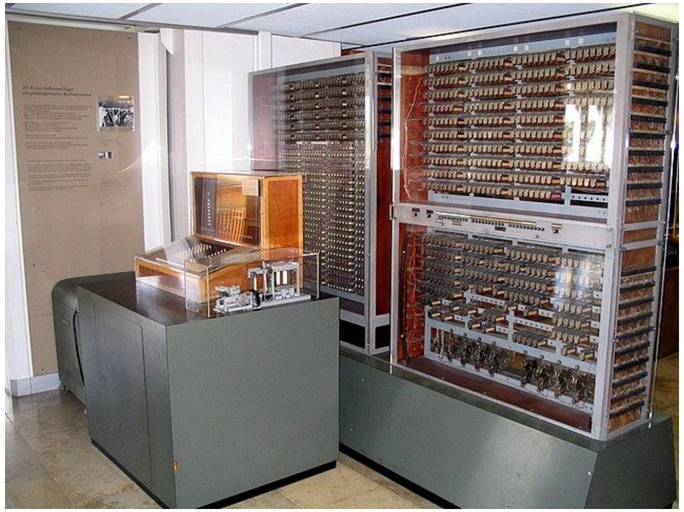
HISTORICAL OVERVIEW : BEGINNINGS

- Gottfried Wilhelm von Leibniz (1646–1716): first mechanical **digital calculator**
- Charles Babbage (1791–1871): **Analytical Engine**, general-purpose program-controlled digital computer (image)
- Ada Lovelace (1815–1852): **first programmer**



HISTORICAL OVERVIEW : PIONEERS

- Konrad Zuse (1910–1995): Z3, first **universally programmable** computer system, 1941
- **Alan Turing** (1912–1954): Key theoretical and practical foundations, e.g. Turing machine



THE INTERNET

- 1969: **ARPANET**, focus on military
- 1990s: **World Wide Web** – Google: 1997
- 2007: iPhone launches era of **smartphones**
- In parallel: Internet of Things/ **Industry 4.0**: interconnected sensors, instruments, and devices networked with industrial applications

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THE AI ERA

- **Metaverse**: Facebook re-named Meta in 2021
- **Artificial intelligence**: ChatGPT launched in 2022
- Anthropic released **Claude Code** in November 2024 (CLI) and rolled out for general use in February 2025
- Google released **Gemma 4** in April 2026.

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- The term describes a society where information production, processing, circulation, and use are central to economic and social life.
- Information becomes a key resource alongside labor, capital, land, and industrial production.
- Computer science provides many of the infrastructures that make this possible.
- The information society is not only about technology; it is about institutions, skills, inequality, and power.

DATA, INFORMATION, KNOWLEDGE

- Data: raw observations or records.
- Information: data interpreted in context.
- Knowledge: information connected to experience, explanation, and action.
- Wisdom/judgment: deciding what should be done and why.

DATAFICATION

- Datafication means turning activities, behaviors, and environments into data that can be stored, analyzed, and acted upon.
- Examples: fitness tracking, learning analytics, online shopping, navigation, smart homes, digital banking.
- Benefits: personalization, efficiency, evidence-based decisions.
  - More access to knowledge and services.
  - New professions and business models.
  - Faster communication and coordination.
  - Personalized education, healthcare, and public services.
- Risks: surveillance, profiling, discrimination, manipulation, loss of autonomy.
  - Privacy loss
  - Platform dependency
  - Misinformation
  - Unequal distribution of benefits

- In the information society, participation requires more than access to a device.
- Citizens need skills for searching, evaluating, creating, protecting, and communicating information.
- Eurostat reported that in 2025, 60% of EU citizens aged 16–74 had at least basic digital skills.
- This is still below the EU Digital Decade 2030 target of 80%.

https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Skills_for_the_digital_age

- Early Web: many independent pages connected by hyperlinks.
- Platform Web: a smaller number of large intermediaries organize communication, commerce, work, and entertainment.
- Examples: Google, Meta, Amazon, Apple, Microsoft, TikTok, Uber, Airbnb.
- Platforms reduce friction but can concentrate power.

WHAT MAKES PLATFORMS POWERFUL?

- Network effects: services become more valuable when more people use them.
- Data advantage: more usage generates more data for optimization.
- Ecosystem control: platforms set rules for developers, sellers, users, and advertisers.
- Lock-in: leaving becomes costly when contacts, history, or workflows are inside the platform.
- Many digital services compete for user attention.
- Recommendation systems decide what content is shown, hidden, amplified, or monetized.
- The information society therefore raises questions about autonomy and public discourse.

Discussion

- Which platform would be hardest for you to stop using?
- What would you lose?
- Who controls the terms?

CASE STUDY, PLATFORM IMPACT

- Choose one platform: Google, Meta, TikTok, Amazon, Microsoft, Apple.
  - What social problem does it solve?
  - What data does it collect?
  - What are its economic incentives?
  - Name two criticisms and one benefit.

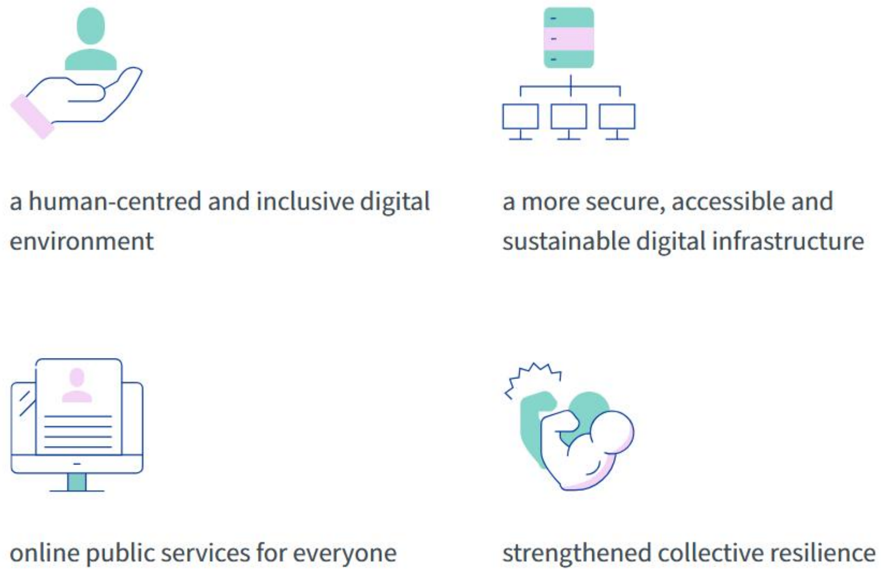
Position yourself: allowed, restricted, or forbidden?

- What is fair assistance and what is misconduct?
- Should students disclose AI use?
- How should assessment change?
- What digital skills should graduates have?

- Digital citizenship means participating responsibly, critically, and safely in digital environments.
- It includes privacy protection, information evaluation, respectful communication, and awareness of algorithmic influence.
- In an information society, citizens need to understand both opportunities and risks of digital systems.
- Computer science education contributes to democratic participation.

EU CONTEXT: DIGITAL DECADE

- The EU Digital Decade sets 2030 targets for skills, infrastructure, businesses, and public services.
- The 2025 report notes progress but also gaps in digital skills, AI adoption, semiconductors, and sovereignty.
- Digital transformation is therefore a political and strategic project, not only a technical one.



https://www.consilium.europa.eu/en/policies/path-to-the-digital-decade-the-eu-s-plan-to-achieve-a-digital-europe-by-2030/

- The EU AI Act applies progressively and introduces obligations based on risk.
- AI literacy and prohibited AI-practice rules became applicable in February 2025.
- General-purpose AI model rules became applicable in August 2025.
- Governance aims to balance innovation, safety, rights, and public trust.
- Why regulation matters for computer science students
  - Computer scientists increasingly work in regulated environments.
  - Design decisions may affect privacy, safety, fairness, accessibility, and transparency.
  - Professional responsibility includes understanding legal and ethical constraints.
  - Regulation is not external to technology; it shapes what systems can and should do.

https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

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SESSION 3

TRANSFER TASK

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Pick a **large private company** working in the field of Information and Communication Technology (ICT)! (E.g. Microsoft, Google, Meta, …)

1. Do your own research to find out about its **impact on the economy** (both in its home country and globally). E.g. **how many people** does it employ, what is its **revenue**, how much does it pay in **taxes**…?
2. What is the impact of this company and its products on **everyday life** and **work life**? How many people use its products?
3. What is the company **criticised** for? List at least 3 points of criticism. Do you agree with the criticism? **Why (not)**?

Prepare a **short report** (in PowerPoint) presenting your findings.

Please present your results.
The results will be discussed in plenary.

1. Who is considered the **first programmer**?
   a) Gottfried Wilhelm von Leibniz
   b) Ada Lovelace
   c) Alan Turing
   d) Konrad Zuse

2. Which percentage of German workers are employed in the **Service sector**?
   a) 55%
   b) 65%
   c) 75%

3. Which is NOT an **organization** connecting individuals to Computer Science?
   a) American Society for Information Science and Technology
   b) Association for Computing Machinery
   c) Institution of Electrical and Electronics Engineers
   d) Association for German Information Engineers

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Bundeszentrale für politische Bildung (2020, November 28). Erwerbstätige nach Wirtschaftssektoren [Employees by economic sector]. https://www.bpb.de/kurz-knapp/zahlen-und-fakten/soziale-situation-in-deutschland/61698/erwerbstaetige-nach-wirtschaftssektoren/
Google LLC - https://gstatic.com/images/branding/googlelogo/svg/googlelogo_clr_74x24px.svg, Gemeinfrei, https://commons.wikimedia.org/w/index.php?curid=42827827
Margaret Sarah Carpenter - UK Government Art Collection website. Original upload was at English wikipedia at en:File:Ada_Lovelace.jpg, Gemeinfrei, https://commons.wikimedia.org/w/index.php?curid=354077
Meta Platforms - about.meta.com/brand/resources/meta/company-brand, Gemeinfrei, https://commons.wikimedia.org/w/index.php?curid=111957412
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